Class 13: Classical Particles and Quantum Particles

In the Drude model, we assumed that the free electrons in a solid behaved in a manner similar that of atoms of an ideal gas. This meant that the electrons followed the Maxwell-Boltzmann statistics. We derived the Maxwell-Boltzmann statistics in the previous class. Let us now briefly consider how the distribution of energy in the system varies as a function of temperature. Figure 13.1 below plots the variation of the Maxwell-Boltzmann distribution as a function of temperature.

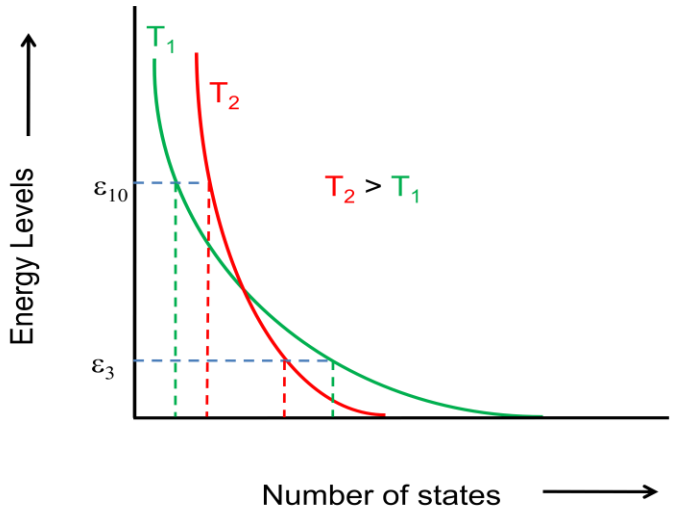


Figure 13.1: Variation of the Maxwell-Boltzmann distribution as a function of temperature. At a higher energy level such as ε_{10}, more states are occupied at the higher temperature T_2, than at the lower temperature T_1. Whereas, at the lower energy level ε_3, less states are occupied at the higher temperature T_2, than at the lower temperature T_1. This layout of energy in the system is consistent with the fact that the overall energy of the system has increased with an increase in temperature.

As indicated in Figure 13.1, at higher energy levels more states are occupied at the higher temperatures, whereas, at the lower energy levels, lesser number of states are occupied at the higher temperatures. This layout of energy in the system is consistent with the fact that the overall energy of the system has increased with an increase in temperature. Figure 13.1 also indicates to us how the layout of energy in the system, changes with the temperature – which is essentially the information that is captured in the specific heat at constant volume of the system. The system in this case being the free electrons in the solid.

Analysis of the predictions of the Drude model have shown that it erroneously predicts the distribution of energy in the system and the specific heat of the system. Therefore we conclude that the Maxwell-Boltzmann distribution is not appropriate for free electrons in a solid. It is very informative to understand why the Maxwell-Boltzmann distribution is inappropriate for electrons in a solid. Such an understanding will enable us to make better decisions on what will be a more appropriate assumption for electrons in a solid. In the

rest of this class, we will therefore closely examine a very fundamental assumption of the Maxwell-Boltzmann distribution and understand the implications of the same as well as recognize the possibilities that exist to modify those assumptions.

The Maxwell-Boltzmann distribution assumes that the particles are what Physicists refer to as „classical' or, in other words, are „identical but distinguishable' particles. It turns out that this is the central assumption that makes the Maxwell-Boltzmann distribution inappropriate for free electrons in a solid. We will therefore examine what is meant by „identical and distinguishable' and also identify other possibilities.

The words „identical but distinguishable', mean something specific when used in the context of Physics. For the longest time, all of the particles and objects we were aware of, were assumed to be „classical' in nature and hence were called classical particles and the associated Physics was called „Classical Physics'. Newton's laws apply to classical particles. Only around the year 1900 did the idea emerge that particles could be considered to behave in a manner other than classical.

Consider any two objects, let us say two balls for example - whether they are identical or not can be decided by comparing their attributes. In Figure 13.2 below, a few different possibilities are considered.

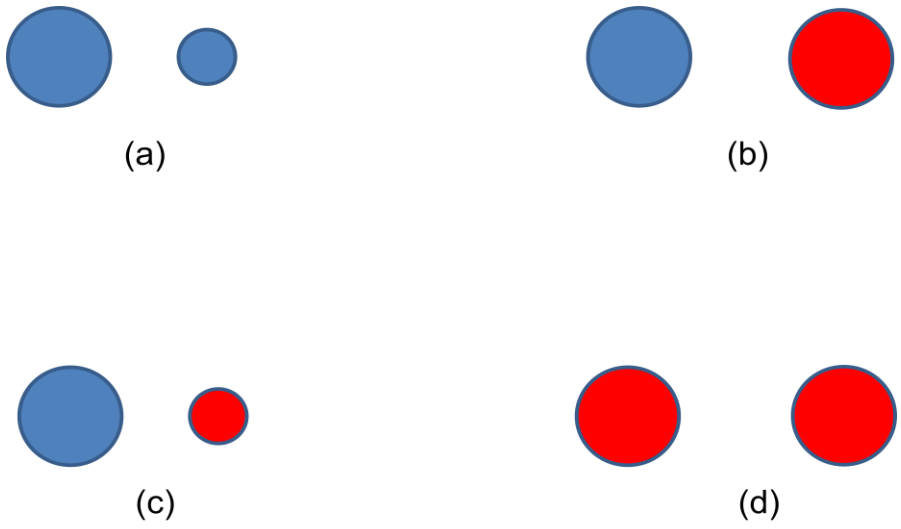


Figure 13.2: Two balls that are a) Not identical in size but identical in color, b) Not identical in color, but identical in size, c) Not identical in size as well as in color, and d) Identical in size as well as color

In the example chosen in Figure 13.2 above it is seen that when two balls are compared based on their attributes, they could differ in size, or color, or both, or could be identical in size as well as color. If the material of manufacture of the balls is the same, then with the same size and color we will have two balls that we can reasonably consider as

„identical'. The question then is whether we can take two such identical balls and still distinguish between them. Figure 13.3 below considers some possibilities.

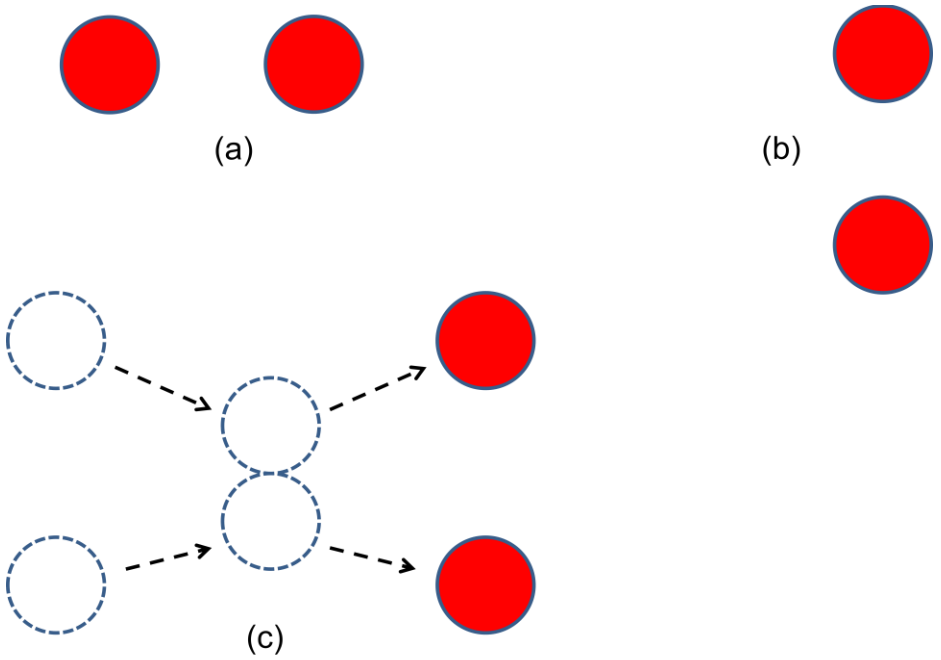


Figure 13.3: Two identical balls that are distinguished based on their position a) Left and right, b) Up and down, and c) Distinguished even when they undergo a collision (The dotted circles represent the position of the balls before and during collision, and the arrows indicate the path of each specific ball).

In the macroscopic world that we are used to we can look at the position of the two balls and state that a specific ball is on the right and a specific ball is on the left. As long as the balls are stationary, which we can easily verify, the ball on the right will remain on the right and the ball on the left will remain on the left for any length of time. In this manner, even though the balls are identical, we are able to distinguish between them. We could use the same sort of reasoning if instead of left and right, one ball was held at a higher position and another at a lower position – again the respective positions will remain undisturbed if there is now relative movement, and we can distinguish between the identical balls. Similarly, if the balls were to move towards each other, collide, and then move apart, as long as we know the initial conditions, we can confidently state which ball is where after the collision. Specifically, there is no possibility that the balls could have mysteriously interchanged their positions without our knowledge. This is the basis of the idea of „identical and distinguishable'. The analysis above may not seem profound at this stage, but its significance will be clear when we consider other possibilities.

While the dimensions of the objects we have discussed above are large, balls which could be several tens of cm in diameter (10^{-1} m), atomic and sub-atomic particles are of much smaller dimensions as indicated in the table 13.1 below:

Particle	**Size scale**
Atoms	10^{-10} m (1Å)
Protons	10^{-15} m
Neutrons	10^{-15} m
Electrons	10^{-18} m

Table 13.1 : The size scale of some atomic and subatomic particles.

The size scales of the particles listed in the table above are several orders of magnitude less than that of the balls discussed so far. The limit of material characterization techniques is only marginally better than the atomic level of 10^{-10} m. As it turns out, when the size scale decreases, the certainty with which we can simultaneously indicate the position of the particle as well as its velocity, also begins to decrease. This is an idea that is central to the field in physics known as „Quantum Mechanics'. It is important to note that this decrease in certainty is **not an experimental limitation** but a phenomenon of nature – something that we will discuss more in the next class.

We can compare subatomic particles in a manner similar to how we compared balls in the earlier discussion. Here the attributes of significance are size and charge. Protons and neutrons are similar in size but differ in their charge. Protons and electrons differ in their charge as well as their size. So these different particles are clearly not identical. The challenge that we face is when we compare two electrons with each other. Two electrons, by definition, have the same size as well as the same charge. They are therefore identical particles. The question is, can we distinguish between two identical particles such as two electrons. In classical physics we make the assumption that we can treat two electron as no different from the two macroscopic balls that we have described earlier, and therefore assume that it will always be possible to distinguish between them. Specifically, in the example of the two balls colliding and moving away, as shown in Figure 13.3c, if each of the balls was actually an electron, classical physics says that the electron at the top of the figure will, with certainty, remain at the top after the collision.

Since the particles are subatomic, and are in motion, the concept of interest is the trajectory of the particle, which is what we have examined in Figure 13.3c. In classical physics, we say that we can keep track of each specific electron by simply keeping track of its trajectory.

Quantum mechanics adopts the position that there is only a probability that a particle is at a given location. This probability could be high or low. If we follow the trajectory of the particle, we can only say that there is a high probability that the particle is where we think it is. There is a definite, and hence non-zero probability, that the particle could be elsewhere too. As particles approach closer to each other, there is an increasing probability that they could interchange positions without our being aware of it. In other words, identical particles could swap positions at anytime, without our knowledge, and hence these identical particles cannot be distinguished from each other. Particles discussed in the context of quantum mechanics, are therefore „identical but indistinguishable'

The central concepts of classical mechanics and quantum mechanics, as discussed above, are highlighted in Figure 13.4 below

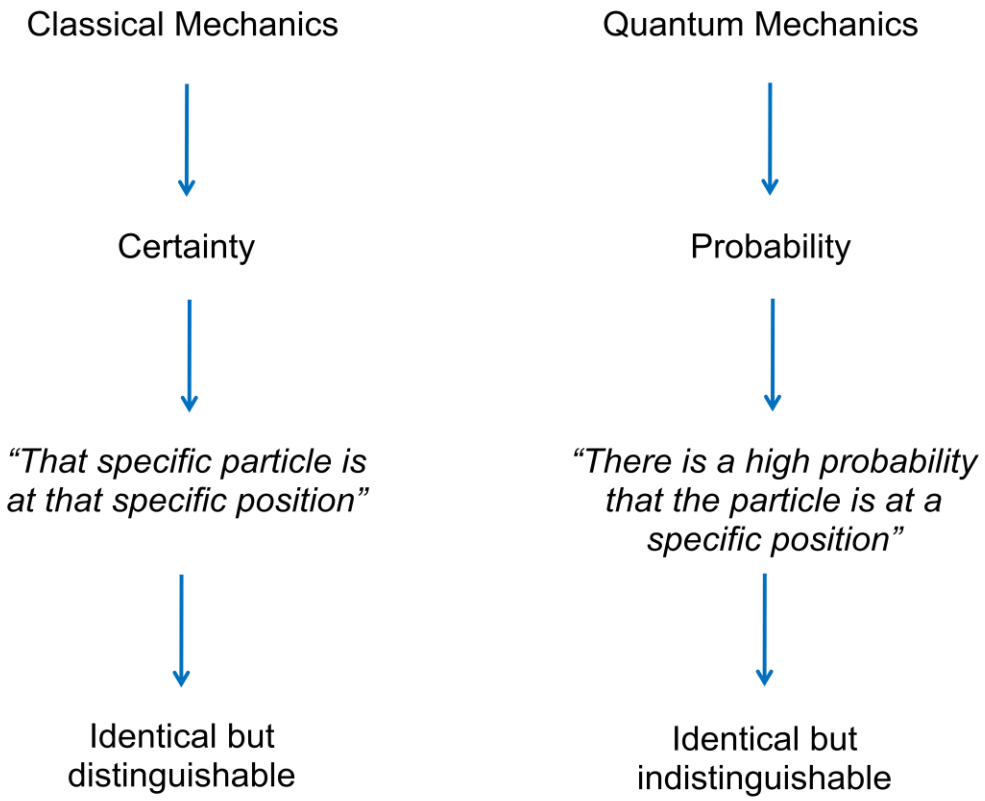


Figure 13.4: The concepts that form the basic ideas of classical mechanics, and how they compare with the basic ideas of quantum mechanics

The distinction made above is very important because it changes the way we count the number of states of a system – an aspect that affects the predictions we make of the system. In classical physics, which uses Maxwell-Boltzmann statistics, when two identical particles, that are occupying two different energy levels, swap positions, the new arrangement is counted as an additional microstate available to the system. In

Quantum mechanics, when two identical particles, that are occupying two different energy levels, swap positions, it is not counted as a new microstate available to the system because the particles may have swapped back without our knowledge anyway.

One of the comments we made earlier is that we will make assumptions about materials and their constituents and then see if the behavior predicted by those assumptions is validated by experimental data. At this time we find that assuming free electrons to be classical particles, i.e. identical but distinguishable, has not led to a satisfactory concurrence with the experimental data. It is therefore of interest to see if the other available approach, which is to treat electrons as quantum mechanical particles, i.e. identical but indistinguishable, leads to a better validation by the available experimental data.

Our present understanding of the way nature functions is that it seems to follow the rules of quantum mechanics. However, quantum mechanical effects become prominent only under specific conditions. We will, at present, just make the assertion that quantum mechanical effects are not prominent in the scheme of ideal gas atoms at room temperature, but are prominent in the scheme of free electrons in a solid. We therefore have to take into account quantum mechanical behavior of electrons, and re-evaluate the predictions that will result.

In the next couple of classes we will look at the history of quantum mechanics. In later classes (class 17 and class 18) we will derive the statistical behavior for a collection of quantum mechanical particles, which is referred to as the Fermi-Dirac statistics in honor of its authors. We will then employ these statistics to the free electrons in the solid and examine the predictions that result.